

Name..... Adm. No..... Sec.....

Put a tick mark (✓) against the correct answer. There is only one correct answer for each question. Each correct answer would fetch 1 mark and no partial credit. Overwriting is not allowed and would attract zero credit.

- (1) The current in a semiconductor diode is often measured by the Shockley equation $I = I_0(e^{5V} - 1)$, where V is the voltage across the diode, I is the resulting current, I_0 is the reverse current. Suppose V follows the density

$$f(x) = \begin{cases} \frac{1}{2}, & 1 \leq v \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Then $E(I) =$ (A) $\frac{I_0}{10}(e^{15} - e^5 - 1)$ (B) $\frac{I_0}{10}(e^{15} - e^5 - 10)$ (C) $\frac{I_0}{10}(e^{15} - e^5)$ (D) $\frac{I_0}{10}(e^{15} - 1)$

- (2) If two numbers are selected in the interval $(0, 1)$ and it is known that the sum of the squares of the number is less than 1, then the probability that the sum of numbers is greater than 0.75 is (take $\pi = 3.1416$):

(A) 0.6419 (B) 0.6374 (C) 0.6512 (D) 0.6674

- (3) The lead concentration in gasoline currently ranges from 0.1 to 0.5 grams per liter. The density f , for the random variable X , the number of grams of lead per liter of gasoline is

$$f(x) = \begin{cases} 12.5x - 1.25, & 0.1 \leq x \leq 0.5, \\ 0, & \text{elsewhere.} \end{cases}$$

Then $P(0.2 \leq X \leq 0.3) =$ (A) 0.75 (B) 0.1875 (C) 0.5 (D) 0.3125

- (4) In a bolt factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the total. Of their output, 5%, 4%, 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machines A, B and C?

(A) $\frac{25}{69}, \frac{28}{69}, \frac{16}{69}$ (B) $\frac{23}{69}, \frac{19}{69}, \frac{17}{69}$ (C) $\frac{27}{69}, \frac{31}{69}, \frac{11}{69}$ (D) $\frac{25}{69}, \frac{27}{69}, \frac{17}{69}$

- (5) Suppose the cumulative distribution function F of X is given by

$$F(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{5}{6}, & 2 \leq x < 3 \\ 1 & 3 \leq x. \end{cases}$$

Then $E(X)$ is (A) $\frac{3}{5}$ (B) $\frac{5}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

- (6) If the probability of seeing a car on the highway in 30 minutes is 0.95, then the probability of seeing a car on the highway in 10 minutes is:

(A) 0.316 (B) 0.618 (C) 0.371 (D) 0.632

[PTO]

- (7) Let X be a random variable with the moment generating function $M(t) = \frac{e^t - 1}{t}$ for all t . Then the variance of X is (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 1 ☒ (D) $\frac{1}{12}$

- (8) Suppose the joint density function of (X, Y) is

$$f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

The marginal density function of X is

- (A) $f(x) = \frac{2}{3}(1+x), 0 \leq x \leq 1$ (B) $f(x) = \frac{1}{6}(2+x), 0 \leq x \leq 2$
(C) $f(x) = \frac{1}{6}(1+2x), 0 \leq x \leq 2$ ☒ (D) $f(x) = 2x^2 + \frac{2}{3}x, 0 \leq x \leq 1$
- (9) For a distribution, Karl Pearson's coefficient of skewness is 0.64, the standard deviation is 13, and the mean is 59.2. Then (mode, median) is:
- ☒ (A) (50.88, 56.42) (B) (56.42, 50.88) (C) (56.42, 57.3) (D) (50.88, 57.3)
- (10) For a distribution, the standard deviation is 10.95. To ensure the distribution is platykurtic according to Pearson's measure of kurtosis, the fourth moment about the mean (μ_4) should satisfy the following condition:

- (A) $\mu_4 < 120$ (B) $\mu_4 < 14400$ ☒ (C) $\mu_4 < 43,200$ (D) $\mu_4 < 1,72,8000$